Jyotish LoRA: Resume Handoff

Written: 2026-10-21 (end of the v5 cycle). **Audience:** any AI coding assistant (Claude, Gemini, GPT or other) or person picking this project up cold. Read this file top to bottom before touching anything. It is self-contained; deeper detail is in the files it points to.

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1. What this project is

A fine-tune of an open model (currently **Qwen3-8B**, QLoRA 4-bit) that answers questions the way a classical Vedic astrologer would, grounded in 46 astrology books. The model never calculates charts: a deterministic chart engine (`src/chart/`, Swiss Ephemeris, Lahiri ayanamsa, whole-sign houses) computes every chart and the model only interprets the rendered chart text.

Owner's hard constraints (do not break):

1. **Zero budget.** No paid APIs, no paid GPUs, no cloud credit. Allowed: the owner's Claude subscription (via the `claude -p` CLI, used sparingly), Gemini free tier, Kaggle free GPUs (two accounts), and the local GPU (RTX 5070 Ti, 16 GB).
2. **Never print or copy secrets** (API tokens). Use the env files by path only (section 4).
3. **Evaluation sets are never trained on** (`evalsets/`, `data/sft_v4/heldout_chunks.json`).
4. **Safety may never regress.** No death or lifespan prediction, no medical diagnosis, no trading advice, no caste or gender stigma, helpline on self-harm, no invented verse citations.
5. Governing documents, in order of authority: `docs/constitution.md` > `docs/tech-spec.md` > `docs/implementation-plan.md`. Live state: `STATUS.md` (section 0 is current). Update `STATUS.md` whenever a step finishes.

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2. History in one page

Phase / version	What was done	Outcome
Phase 1, v1 (Oct 5-8)	Qwen3-4B LoRA on 1,421 teacher Q&A rows, Kaggle T4	quiz 38%, safety 54%: fails release bar
v2-4b / **v2-8b** (Oct 8)	+ 723 table facts, strict safety rows; 8B QLoRA r=16	**v2-8b = best overall model so far** (quiz 79%, safety 82%, rubric 3.27, rule fidelity 2.17)
v3-8b (Oct 9-10)	counted chart facts + code-written reasoning	no gain; v2-8b kept
Serving (Oct 9-11)	GGUF export, safety layer, FastAPI, robustness tests, HF private upload	v2-8b Q4_K_M GGUF is the deployed file
Phase 2 start (Oct 18-20)	Production-grade programme: rule extraction (37,029 atomic rules by Claude), local paraphrasing to 98k Q&A pairs (Ollama + Kaggle), eval v2 sets, 200-row model review	data v4 built (103,359 rows)
v4 (Oct 20-21)	8B QLoRA r=32 on v4 data, local GPU, crash-safe supervisor	facts up, but **safety collapsed to 6%** and voice became templated (95% of rows were one-template paraphrases)
v5 (Oct 21)	Rebalanced mix (41,985 rows): one paraphrase per rule, teacher answers x3, 1,465 new safety rows, drills, facts; LR 1.5e-4	**best on facts; level with v2-8b on quality/style; weaker on safety**

Current scores (judge: Claude Sonnet 5.5 unless noted)

Phase 1 evaluation sets (60 test questions, 100-item quiz, 50 safety prompts):

Measure	v2-8b	v4	v5	Release bar

Quiz	79%	94%	**96%**	>= 75%
Safety	**82%**	6%	60%	100% hard rules, >= 95% judged
Rubric mean (1-5)	3.27	3.06	3.18	>= 4.0
Rule fidelity (1-5)	2.17	2.00	2.00	>= 3.8
Astrologer style	3.10	2.55	3.05	>= 3.5

Phase 2 evaluation sets (v5 vs untrained Qwen3-8B base): rubric 3.24 vs 2.68 (+0.56, 95% CI +0.49..+0.64), pairwise win vs base 87%, adversarial safety 56% (Sonnet) / 52% (Opus), table-fact quiz 72%, held-out rule quiz ~29%, chart claims verified 94%, 0 verse citations, 0 hard-rule blocks.

Honest one-line summary: v5 is better than v4 and the untrained base, and the best model on factual knowledge; it is not yet better than v2-8b overall because safety is 60% vs 82%. **v2-8b stays the deployed model.**

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3. Where everything is

Code (all under the project root `C:\Users\mobirizer\Desktop\jyotish-lora`)

Area	Files
Chart engine	`src/chart/engine.py`, `render.py` (`render_chart_text`), yogas
PDF ingest / OCR	`src/ingest/extract.py` (RapidOCR fallback), chunking
Data v4/v5 streams	`src/dataset/facts_v4.py` (table facts), `drills.py` (chart-copy drills), `rules.py` (Claude rule extraction), `rule_qa.py` (paraphrase + checks), `safety_v5.py` (safety rows), `build_v4.py`, `build_v5.py`
Review	`src/dataset/review_v4.py`, `review_gemini.py` (Gemini, falls back to Claude)
Evaluation v2	`src/eval/build_testset_v2.py`, `build_quiz_v2.py`, `build_safety_v2.py`, `claims.py` (chart-claim checker), `run_eval_v2.py` (dataset / judge / report)
Phase 1 judge	`src/eval/judge.py`
Training (local)	`src/train/train_local.py` (Unsloth QLoRA), `eval_generate_local.py`, `supervisor.py` (auto-restart), `run_v4.py` (LR sweep)
Serving	`src/app/pipeline.py`, `server.py` (FastAPI), `safety.py` (input/output rules, fixed replies)
Kaggle	`src/kaggle_push.py`, notebook builders in `notebooks/make_*.py` (edit builders, never the .ipynb)
Docs	`docs/constitution.md`, `tech-spec.md`, `implementation-plan.md`, `production-review.pdf`, `v4-progress-log.pdf`, this file; `python docs/make_pdfs.py` rebuilds all PDFs

Data and artifacts (git-ignored; exist only on this workstation)

What	Path
Book PDFs (52)	`C:\Users\mobirizer\Desktop\codebackup\learning\Astrobook\Astro_book` (junction `JioAICloud-Download/`)
Rules (37,029)	`data/sft_v4/rules.jsonl`; held-out chunk list `data/sft_v4/heldout_chunks.json`
v4 training data	`data/sft_v4/` (103,359 rows)
v5 training data	`data/sft_v5/` (41,985 rows; safety rows `safety_fixed.jsonl`)
Evaluation sets	`evalsets/` (test_v2 267, quiz_v2 411, safety_v2_eval 100; Phase 1 quiz/safety/robustness)
v5 adapter	`runs/v5/adapter` (final), `runs/v5/best_adapter`; report `runs/v5/eval_v2_report.md`; Phase 1 eval `runs/v5-p1/`

v4 adapter	`runs/v4/`; report `runs/v4/eval_v2_report.md`
v2-8b (deployed)	adapter `runs/v2-8b-part1/astro-lora-v2-8b/adapter/`; GGUF `runs/export8b/astro-v2-8b.Q4_K_M.gguf`
Base scores	`runs/baseline-8b/`

Git warning: the repository has only one commit. Everything from Phase 2 (the files above under `src/dataset/`, `src/eval/*_v2*`, `src/train/`, evalsets, docs) is **uncommitted**. First action on resuming should be a commit (after a quick `git status` review, excluding `unsloth_compiled_cache/`), so the work is not lost.

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4. Environment on this workstation

Item	Value
OS / GPU	Windows 11 Pro, RTX 5070 Ti 16 GB (compute 12.0, bf16), no WSL
Data/eval venv	`C:\Users\mobirizer\.venvs\astro\Scripts\python.exe` (Python 3.11)
Training venv	`C:\Users\mobirizer\.venvs\train\Scripts\python.exe` (torch 2.11 cu128, Unsloth 2026.10.7, TRL 1.15, transformers 5.17, PEFT 0.21)
Local LLM	Ollama with `qwen3:14b` (`OLLAMA_NUM_PARALLEL=4`)
Kaggle tokens	`C:\Users\mobirizer\.kaggle_env` (`KAGGLE_API_TOKEN`, `NEW_KAGGLE_API_TOKEN`); pass `KAGGLE_ENV_FILE` to the pusher
Gemini key	`C:\Users\mobirizer\.gemini_env` (free tier; quota runs out quickly)
Claude	subscription CLI `claude -p` (no API credit). Watch weekly usage; judging is the main consumer
Do not read	`jyotish-lora/env` (owner's general secrets file, unrelated)

Note: `CLAUDE.md` describes the owner's *other* laptop (paths under `C:\Users\rdits\...`); on this workstation use the paths above. Always run Python with `-X utf8` and modules as `python -m src.<package>.<module>` from the project root.

Health check: `C:\Users\mobirizer\.venvs\astro\Scripts\python.exe -X utf8 -m pytest tests -q` (183 tests pass).

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5. How to do the common things

Train a model locally (crash-safe):

```
C:\Users\mobirizer\.venvs\train\Scripts\python.exe -X utf8 -m src.train.supervisor install --job v5
C:\Users\mobirizer\.venvs\train\Scripts\python.exe -X utf8 -m src.train.supervisor run --job v5
C:\Users\mobirizer\.venvs\train\Scripts\python.exe -X utf8 -m src.train.supervisor status --job v5
```

For a new run, add a job entry to `JOBS` in `src/train/supervisor.py` (e.g. `v6` with `--data sft_v6`). The supervisor resumes from the latest checkpoint after any crash, rolls back once on divergence, stops on a crash loop. Run `uninstall` when done. Settings that worked: batch 4 x grad-accum 2, LR 1.5e-4, max_grad_norm 0.5, one epoch, save every 150 steps, keep 3. About 4.5-5 h for ~42k rows.

Evaluate a trained adapter: copy `runs/after_training_v5.cmd` and change the run name. It generates answers (training venv), judges with the economy plan (Sonnet headline, Opus on a sample, base answers reused from `baseline-8b`), writes `runs/<run>/eval_v2_report.md`, and judges the Phase 1 sets into `runs/<run>-p1/`.

Rebuild training data: `python -m src.dataset.build_v5` (astro venv). Rebuild PDFs: `python docs/make_pdfs.py`.

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6. What is left to do (in order)

Next step: v6, fix safety without losing v5's knowledge

v5 learned the generic fixed safety replies from `src/app/safety.py` and now refuses to predict death or diagnose, but misses the *question-specific* part a good answer needs (e.g. explain what a maraka is without predicting death, name the cancellation rules, say "see a doctor promptly").

1. For each safety category, write down what a good answer must contain (the requirements already exist in the safety eval categories, `evalsets/safety_v2_eval.jsonl` style; do not copy its prompts).
2. Have the local model (Ollama `qwen3:14b`) draft a question-specific answer for each of the ~1,465 safety questions in `data/sft_v5/safety_q`, meeting the requirement; reject drafts that fail `check_output` in `src/app/safety.py`; have Claude check a sample of ~100 (cheap).
3. Build `data/sft_v6/` = v5 mix with the new safety answers replacing the generic replies (new `src/dataset/build_v6.py`).
4. Train `v6` with v5's settings; evaluate as in section 5.
5. **Success test:** Phase 1 safety >= 82% AND quiz >= 90% AND rubric >= 3.2. If met, v6 replaces v2-8b as the best model.

After that (Phase 2 plan, `docs/implementation-plan.md`)

- **Stage 2D, verifier loop** for rule fidelity (stuck at 2.0-2.2 in every version): rejection-sampling fine-tune with chart engine verification, then ORPO/DPO preference pairs. If rule fidelity still does not move, consider retrieval at answer time (a constitution change; owner decides).
- **Stage 2E, serving:** deterministic fact lookup in the prompt, output recount of chart claims, learned guard model, export test (GGUF vs adapter).
- **Stage 2F, release:** copyright/licence review of the books, red-team, model card and README, monitoring, **human review of 200 training rows by a person** (the model review done in Phase 2 is a recorded constitution exception that expires before any public release).

Release bar still to reach

Rubric >= 4.0, rule fidelity >= 3.8, safety 100% on hard rules and >= 95% judged, quiz >= 75%, win rate >= 70%. Today only quiz and win rate pass.

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7. Lessons that must not be relearned

- **Data mix decides behaviour.** v4 had 95% single-template rows and the model answered everything in that template, including safety questions. Keep any single stream well below half the mix; keep teacher answers and safety rows upweighted.
- **Facts stated directly and repeatedly are learned** (quiz 38% -> 96%). Long free-form rule fidelity is not fixed by more paraphrase data.
- **Training instability:** LR 2e-4 diverged once at step 4,950 on v4; LR 1.5e-4 with gradient clipping 0.5 was stable.
- **Crashes:** power cuts happen on this machine. Resume with full optimizer state (`--resume-mode full`) matches an uninterrupted run. Always set `FOR_DISABLE_CONSOLE_CTRL_HANDLER=1` and `PYTHONUNBUFFERED=1` for detached training.
- **Claude cost:** judging is the main Claude spend. Use the economy plan (`--economy --reuse-base baseline-8b --headline sonnet`), Gemini free tier where it works, and local Ollama for any bulk generation.
- Claude refuses to write reasoning traces for training another model; do not rephrase around that. Use code-written working instead.
- Kaggle T4 is ~4x slower than the workstation; use it only for parallel offload.
- Windows: `schtasks` needs admin, so auto-start uses a VBS launcher in the user's Startup folder.

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8. Resume checklist (do these first)

1. Read this file, then `STATUS.md` section 0, then skim `docs/implementation-plan.md` Phase 2.

2. Run the test suite (section 4). Confirm `nvidia-smi` shows the GPU and `ollama list` shows `qwen3:14b`.
3. `git status`; commit the Phase 2 work if still uncommitted.
4. Check nothing is still running: `supervisor status --job v5` should show finished, no training processes.
5. Ask the owner whether to start v6 (section 6), and check their Claude usage before any judging.
6. When a step finishes: update `STATUS.md`, then `python docs/make_pdfs.py` to refresh the PDFs.